Back **KD and range trees**
Graded Quiz • 30 min

English Due Sep 8, 11:59 PM IST

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KD and range trees

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1. In the KD-tree queries we have used rectangular regions as the query region. Can the query algorithm also be used with triangular regions? **1 / 1 point**

Correct

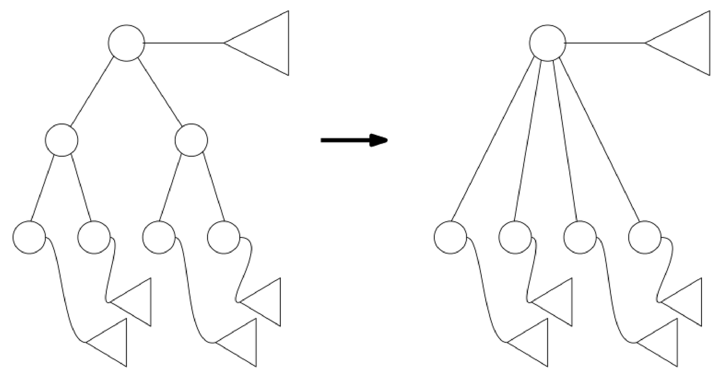
2. Suppose we have a collection of n points in the plane that all lie on the diagonal $y = x$. What is the **worst case** KD-tree query time of this input? Is there a simpler linear-size data structure achieving the same or better running time? **1 / 1 point**

Correct

3. Suppose we have a collection of n points in the plane that all lie on the diagonal $y = x$. What is the **best case** KD-tree query time of this input? **1 / 1 point**

Correct

4. We have seen that the two-dimensional range tree on a set of n points requires $O(n \log n)$ storage. One could bring down the storage requirements by storing associated structures only with a subset of the nodes in the main tree. Suppose we only store associated structures at certain depths in the tree, this is equivalent to storing a range tree with a higher degree at each node. For example, if we store associated structures every other level, then this is equivalent to storing a tree with degree 4 at every node where we store an associated structure at every internal node. We basically skip the intermediate node that did not have an associated structure, as shown in the image below. Suppose we use this technique to create a range tree where each node has degree k (except for the leaves), what happens to the storage requirement and the query time? Assume that we do not use fractional cascading. **1 / 1 point**



Correct

5. Which of the following statements about **two-dimensional** KD-trees and range trees are correct? **1 / 1 point**

Correct